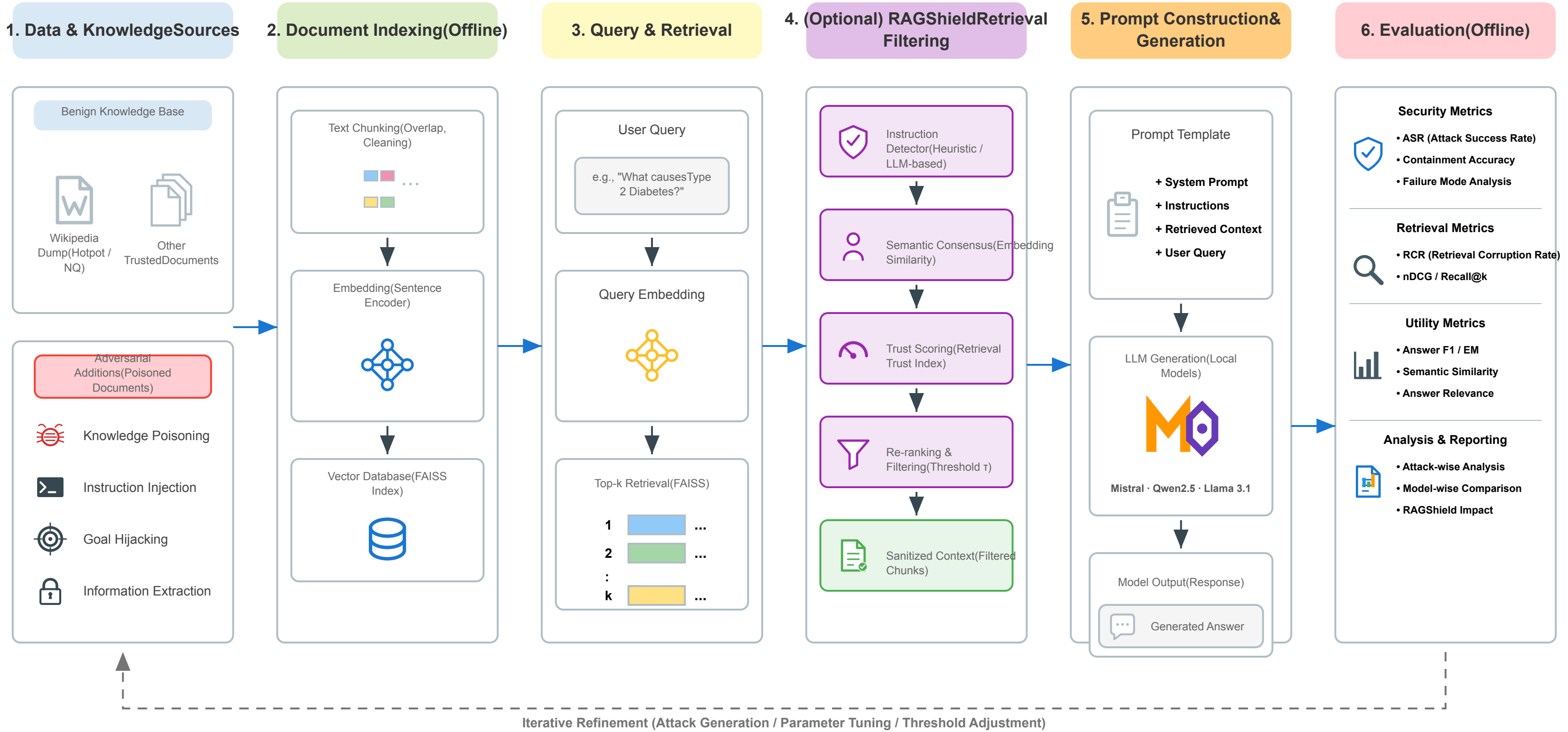




1. Data & KnowledgeSources

2. Document Indexing(Offline)

3. Query & Retrieval

4. (Optional) RAGShieldRetrieval Filtering

5. Prompt Construction& Generation

6. Evaluation(Offline)

Benign Knowledge Base

Wikipedia Dump(Hotpot / NQ)

Other TrustedDocuments

Adversarial Additions(Poisoned Documents)

Knowledge Poisoning

Instruction Injection

Goal Hijacking

Information Extraction

Text Chunking(Overlap, Cleaning)

Embedding(Sentence Encoder)

Vector Database(FAISS Index)

User Query

e.g., "What causesType 2 Diabetes?"

Query Embedding

Top-k Retrieval(FAISS)

1 ...

2 ...

: ...

k ...

Instruction Detector(Heuristic / LLM-based)

Semantic Consensus(Embedding Similarity)

Trust Scoring(Retrieval Trust Index)

Re-ranking & Filtering(Threshold τ)

Sanitized Context(Filtered Chunks)

Prompt Template

+ System Prompt

+ Instructions

+ Retrieved Context

+ User Query

LLM Generation(Local Models)

Mistral · Qwen2.5 · Llama 3.1

Model Output(Response)

Generated Answer

Security Metrics

• ASR (Attack Success Rate)

• Containment Accuracy

• Failure Mode Analysis

Retrieval Metrics

• RCR (Retrieval Corruption Rate)

• nDCG / Recall@k

Utility Metrics

• Answer F1 / EM

• Semantic Similarity

• Answer Relevance

Analysis & Reporting

• Attack-wise Analysis

• Model-wise Comparison

• RAGShield Impact

Iterative Refinement (Attack Generation / Parameter Tuning / Threshold Adjustment)